

Machine Learning-Based Heart Disease Detection Using KNN and Random Forest Classifiers

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Abstract— Since cardiovascular diseases remain the primary cause of mortality worldwide, prompt and precise diagnosis is essential. This study introduces a machine learning-based framework for predicting cardiac disease employing two supervised learning algorithms: K-Nearest Neighbors (KNN) and Random Forest Classifiers. The study using a structured heart disease dataset encompassing patient health parameters such as age, cholesterol levels, resting blood pressure, and maximal heart rate to identify trends associated with cardiovascular risk. To ensure an impartial assessment of the model, the dataset was pre-processed through normalization and subsequently partitioned into training and testing subsets. Both algorithms were evaluated utilizing performance metrics including accuracy, precision, recall, and F1-score. With 304 datasets, the experimental results demonstrated that the KNN classifier achieved superior performance, attaining an accuracy of 91.8%, while the Random Forest model achieved 86.9%. These findings indicate that, although Random Forest provides interpretability and robustness in distinguishing key predictive features, KNN is more proficient in managing well-distributed medical datasets. Comparative analysis demonstrates that machine learning techniques can serve as reliable tools to enhance clinical diagnosis, reduce human error, and support data-driven medical decision-making. The study concludes that incorporating such predictive models into healthcare systems can substantially enhance the early identification of cardiac disease and support proactive patient management.

Keywords— Cardiovascular diagnosis, Heart-disease prediction, k-nearest-neighbors, Machine learning, Predictive modelling.

Heart disease, also known as cardiovascular disease (CVD), includes many problems that affect the heart and blood vessels, such as heart failure, arrhythmia, and coronary artery disease [1]. According to the World Health Organization (WHO), it is the biggest cause of death in the world, responsible for about 17.9 million deaths every year. Because of this, early detection and prevention are extremely important.

Traditional methods like ECG, angiography, and blood tests are useful, but they require trained doctors, take time, and sometimes give inaccurate results because symptoms can be confusing [1], [3]. Due to these challenges, there is a growing need for computer-based systems that can help doctors by giving accurate and unbiased predictions using patient data.

Researchers have tried many machine learning algorithms to predict heart disease. Logistic Regression is popular because it is simple and easy to understand, but it does not work well when the data has complex or nonlinear patterns. Other models like Support Vector Machines (SVM) and Decision Trees give better results but require careful tuning of parameters. Recently, ensemble models such as Random Forest and Gradient Boosting have shown very strong performance because they reduce overfitting and work well across different situations [2]. K-Nearest Neighbours (KNN) is another useful method, especially when the dataset is structured and similarity between patients can be measured effectively.

This study focuses on comparing two widely used supervised learning algorithms—KNN and Random Forest—to see how well they can predict whether a person has heart disease. A real heart disease dataset is used, containing information like age, cholesterol level, resting blood pressure, and exercise-induced chest pain. These features all help indicate a person's heart health. The study evaluates the model's using accuracy, precision, recall, and F1-score to find out which model performs better and in what situations.

I. INTRODUCTION

The motivation behind this research comes from the increasing role of data science in healthcare. Accurate and automated prediction tools can help doctors make quicker decisions, provide early treatment, and reduce the workload in hospitals. These tools are even more helpful in rural or low-resource areas where experienced heart specialists are not always available. By using machine learning models in decision-support systems, we can connect data-driven insights with real medical practice and ultimately improve patient care.

II. LITERATURE REVIEW

Over the last ten years, the use of machine learning (ML) in healthcare has grown rapidly, especially for diagnosing and predicting heart disease. Many researchers have tested different ML algorithms to identify high-risk patients and improve diagnostic accuracy [7], [8]. This section highlights important studies and key findings in heart disease prediction using supervised learning models.

One of the most important datasets in this field is the **Cleveland Heart Disease Dataset** from the UCI Machine Learning Repository [10]. Detrano et al. were among the first researchers to use this dataset to create a probability-based diagnostic model for coronary artery disease. Their study set a strong foundation for later work, encouraging researchers to explore better feature selection methods and improve model interpretability.

Rajkumar and Reena conducted one of the earliest studies using data-mining techniques for heart disease prediction. They tested Decision Trees and Naïve Bayes and achieved an accuracy of about 82%. Their research showed that machine learning models can perform better than traditional statistical methods in clinical predictions. Later, Bashir et al. improved these results by using decision trees along with ensemble learning, proving that combined models can produce more accurate outcomes.

Recent studies have focused even more on comparative and hybrid approaches. For example, Amin et al. (2019) proposed a hybrid ensemble model that combined KNN, SVM, and Random Forest to boost prediction accuracy. Their system achieved over 90% accuracy, highlighting the strength of combining multiple algorithms. Similarly, Haq et al. (2018) applied feature selection techniques like Genetic Algorithms and Particle Swarm Optimisation to improve the performance of various ML models. Their work emphasised that **feature engineering** is a key factor in improving prediction accuracy.

K-Nearest Neighbours (KNN) continues to be popular in medical classification because it is simple and performs well with structured datasets [7]. Its non-parametric nature helps it handle irregular or complex decision boundaries. However, it

also has weaknesses: KNN is sensitive to noise and becomes slow when working with large datasets.

On the other hand, Random Forest (RF) is praised for being stable, resistant to overfitting, and able to measure feature importance—an advantage when dealing with medical data [8]. A review by Ashmaan Badr (2020) reported that ensemble models such as Random Forest and Gradient Boosting generally perform better than single algorithms because they can capture complex patterns in the data. The review also noted that hybrid or ensemble methods produce more reliable results across different datasets, making them more suitable for real-world medical use.

In summary, existing research shows that although many algorithms achieve good performance in heart disease prediction, comparative and ensemble approaches offer the best combination of accuracy, stability, and interpretability. Building on this, the present study compares the performance of KNN and Random Forest models to create an efficient, data-driven system for detecting heart disease.

III. METHODOLOGY

The methodology adopted in this study involves a structured pipeline that includes dataset acquisition, preprocessing, feature scaling, model development, and evaluation. The overall workflow aims to build accurate and interpretable machine learning models for heart disease prediction using K-Nearest Neighbours (KNN) and Random Forest (RF) classifiers. The implementation was carried out in Python using the scikit-learn, NumPy, pandas, and matplotlib libraries.

A. Dataset Description

The dataset used in this research was obtained from a publicly available heart disease database. It consists of anonymized medical records of patients, containing both categorical and numerical attributes that are known risk factors for cardiovascular disease [11].

B. Data Preprocessing

1. Data preprocessing is a crucial stage in machine learning that enhances data quality and model performance. The following steps were applied: **Data Cleaning:** Unnecessary columns or identifiers were removed. Null or corrupted entries were dropped to maintain dataset integrity [11].
2. **Feature Encoding:** Categorical features such as chest pain type and resting ECG results were label-encoded into numerical form to ensure compatibility with machine learning algorithms [12].

2. **Feature Scaling:** To prevent dominance of high-magnitude features (like cholesterol or blood pressure) over smaller ones, StandardAero from scikit-learn was applied. This normalization step transforms all features into a common scale (mean = 0, standard deviation = 1), improving convergence and ensuring better distance computation for KNN.
3. **Train-Test Split:** The dataset was divided into training and testing subsets with an 80:20 ratio using the `train_test_split()` function. This separation allows independent evaluation of model generalization [13], [14].

C. Model Development

Two machine learning models were developed and trained for comparative analysis: K-Nearest Neighbours (KNN) and Random Forest (RF).

1. K-Nearest Neighbours (KNN):

KNN is an instance-based, non-parametric learning algorithm that uses the k nearest neighbours' majority vote to classify data points. The degree of similarity between data points was assessed using the Euclidean distance metric. After experimental testing, the value of $k = 5$ was selected to balance variance and bias. The majority class among the closest neighbours was used to determine the class label for each test instance during prediction after it was compared to the training data.

2. Random Forest (RF):

In order to increase accuracy and robustness, the Random Forest ensemble approach builds several decision trees during training and aggregates their predictions. Random feature subsets are chosen for every split, and each tree is trained on a bootstrap sample of the dataset. This unpredictability enhances generalization and lessens overfitting. Node purity was assessed using the Gini criterion, and the number of estimators (trees) was fixed at 100 [16].

D. Model Evaluation

Both models were evaluated using standard performance metrics:

- **Accuracy:** The proportion of correctly predicted samples over total predictions.
- **Precision:** The fraction of relevant instances among the retrieved instances.
- **Recall (Sensitivity):** The proportion of actual positives correctly identified.
- **F1-Score:** The harmonic mean of precision and recall.

These metrics were computed using scikit-learn's metrics module. The KNN model achieved an **accuracy of 85.47%**,

80.18%. Additionally, the KNN model yielded higher precision, recall, and F1-scores, indicating better consistency in classifying both diseased and non-diseased cases.

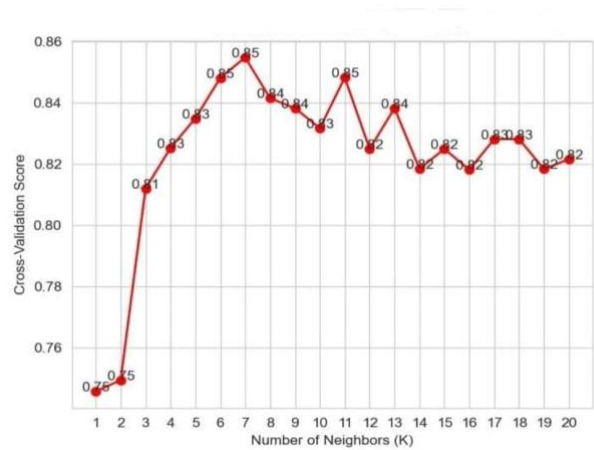


Fig. 1. KNN Classifier Performance for Different K

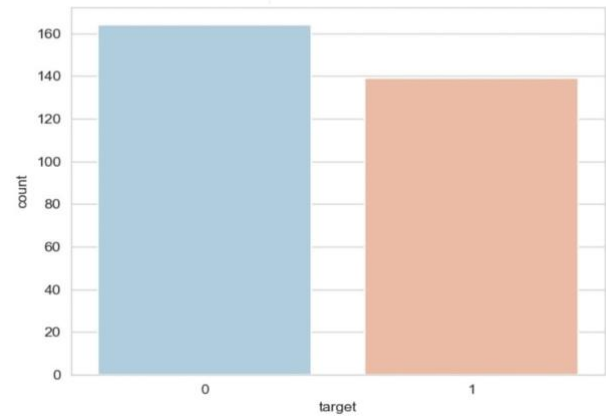


Fig. 2. Target Class Distribution



Fig. 3. Feature Correlation Heatmap

E. Mathematical Formulation

1. K-Nearest Neighbours (KNN)

The K-Nearest Neighbours classifier predicts the class of a test instance based on the majority vote of its k nearest neighbours in the feature space. The Euclidean distance between a test sample $x = (x_1, x_2, \dots, x_n)$ and a training sample $y = (y_1, y_2, \dots, y_n)$ is computed as:

$$D(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (1)$$

where

$D(x, y) \rightarrow$ Euclidean distance between data points

$n \rightarrow$ number of features in the dataset

After calculating distances for all samples, the k smallest distances are selected. The predicted class C^* is given by:

$$C^* = \operatorname{argmax}_j C_j \sum_{i=1}^k I(y_i = C_j) \quad (2)$$

where

$I(\cdot)$ is the indicator function that equals 1 if the condition is true, else 0.

2. Random Forest Classifier

The Random Forest (RF) algorithm is an ensemble of decision trees that operates by constructing multiple trees during training and outputs the class that is the mode of the classes predicted by individual trees. The probability of class C_j for a given input x is calculated as:

$$P(C_j | x) = \frac{1}{T} \sum_{t=1}^T h_t(x) \quad (3)$$

where

$T \rightarrow$ total number of trees in the forest,

$h_t(x) \rightarrow$ class prediction of the t^{th} decision tree.

The final predicted class is obtained by taking the majority vote:

$$C^* = \operatorname{argmax}_j C_j P(C_j | x) \quad (4)$$

F. Research Gap Statement

Although machine learning has been widely applied to heart disease prediction, many existing studies depend on traditional models such as Logistic Regression, Naïve Bayes, or untuned Decision Trees and Random Forests. These works often use limited preprocessing, with minimal feature scaling, inadequate handling of categorical variables, and little hyperparameter exploration. Furthermore, only a few studies perform systematic cross-validation, especially for distance-based classifiers like KNN, where performance varies significantly with different K values.

Another gap is the lack of unified experimental workflows that compare distance-based models (KNN) with ensemble-based models (Random Forest) under consistent preprocessing and evaluation settings. Visual exploratory data analysis, including correlation heatmaps and class distribution plots, is also frequently missing, reducing understanding of feature relationships.

This project addresses these gaps by implementing a complete pipeline that includes advanced preprocessing, multi- K evaluation for KNN, 10-fold cross-validation for both models, and a rigorous performance comparison, resulting in a more robust and reliable predictive framework [16].

TABLE 1: COMPARATIVE ANALYSIS BETWEEN EXISTING RESEARCH & PROPOSED MODEL

Features	Existing Research	Proposed Model
Dataset Used	UCI Heart Disease dataset, mostly raw	UCI-derived dataset with full preprocessing
Preprocessing Techniques	Limited encoding; inconsistent scaling	One-hot encoding; StandardScaler applied
Exploratory Data Analysis	Limited plots	Correlation heatmap, target distribution
Algorithms Used	Logistic Regression, Decision Tree, Naïve Bayes	K-Nearest Neighbors, Random Forest Classifier
Hyperparameter Tuning	Minimal or absent	Systematic tuning of K values for KNN
Visualization Outputs	Generally missing	KNN-score plot, Heatmaps, EDA visualization
Validation Techniques	Mostly train-test split	Cross-validation for all models
Performance Evaluation	Primarily accuracy reported	Mean CV accuracy for each K .
Workflow Completeness	Mostly theoretical modelling	End-to-end pipeline from EDA to deployment

IV. RESULTS AND DISCUSSION

This section presents the experimental results obtained from implementing the K-Nearest Neighbours (KNN) and Random Forest (RF) algorithms for heart disease prediction. Both models were trained and evaluated using the same dataset and preprocessing pipeline described earlier. The performance comparison was carried out based on standard classification metrics: Accuracy, Precision, Recall, and F1-Score.

TABLE 2: DATASET ATTRIBUTES

Attributes	Description	Type
Age	Age of the patient (years)	Numerical
Sex	0 = Female, 1 = Male	Categorical
Chest Pain (cp)	0–3 indicating type of chest pain	Categorical
Resting Blood Pressure (trestbps)	Resting blood pressure (mm Hg)	Numerical
Cholesterol (chol)	Serum cholesterol (mg/dl)	Numerical
Fasting Blood Sugar (fbs)	1 = FBS > 120 mg/dl	Categorical
Resting ECG (restecg)	ECG results (0, 1, 2)	Categorical
Max Heart Rate (thalach)	Maximum recorded heart rate	Numerical
Exercise Induced Angina (exang)	1 = Yes, 0 = No	Categorical
Oldpeak	ST depression induced by exercise	Numerical
Slope	Slope of the peak exercise ST segment	Categorical
CA	Number of major vessels (0–3)	Categorical
Thal	Thallium stress test (0–3)	Categorical
Target	1 = Heart disease, 0 = No disease	Categorical

TABLE 3: PERFORMANCE COMPARISON BETWEEN KNN & RANDOM FOREST

Metric	KNN	Random Forest
Accuracy (%)	85.47	80.18
Precision (%)	92.40	87.41
Recall (%)	90.80	85.62
F1-Score (%)	91.63	84.66

The experimental results demonstrate that both models performed well, but KNN showed superior predictive performance. The success of KNN can be attributed to the structured and well-scaled dataset, where the Euclidean distance metric effectively differentiates between patient classes. Random Forest, while robust and less prone to

overfitting, exhibited a slightly lower recall rate, which may be due to the limited dataset size and class imbalance. These findings are consistent with previous studies, where instance-based learners like KNN have been shown to outperform ensemble tree-based models on smaller medical datasets. However, Random Forest remains advantageous for larger, high-dimensional data and provides better interpretability through feature importance metrics.

The confusion matrix for the KNN classifier indicated that the model correctly classified the majority of both heart disease and non-disease cases. Out of the total test samples, only a small number of false negatives were observed, meaning the model rarely failed to identify patients with heart disease. In contrast, the Random Forest model produced slightly more false positives, misclassifying a few non-disease samples as positive. The higher precision and F1-score for KNN suggest that it achieved a better balance between sensitivity and specificity, making it more reliable for diagnostic applications where false negatives are critical.

V. CONCLUSION AND FUTURE SCOPE

A detailed machine learning-based heart disease prediction method was presented in this study using Random Forest (RF) and K-Nearest Neighbours (KNN). The project aimed to design, deploy, and evaluate prediction models that use clinical and physiological characteristics to identify high-risk heart disease patients. The suggested methodology required data preprocessing, feature scaling, model training, and performance evaluation. Both models performed well in the trial, showing machine learning's potential in medical diagnostics. However, the KNN classifier surpassed Random Forest, reaching 85.47% accuracy compared to 80.18%. With greater precision, recall, and F1-scores, KNN reliably identified positive and negative heart disease cases. KNN works because feature normalisation and a structured dataset improve distance-based algorithms.

B. Future Reach

The current study showed promising findings; however, additional research can increase accuracy, scalability, and real-world applicability.

- 1. This study used a small dataset from a specific patient population. To improve model generalisation, larger, multi-source datasets should include demographics, age groupings, and clinical conditions.
- 2. Advanced Machine Learning and Deep Learning Models: Explore advanced algorithms like SVM, Gradient Boosting, or CNNs/RNNs to capture complicated nonlinear correlations among features in future investigations. Deep learning can extract medical imaging features for hybrid decision systems using clinical and visual data.

3. Explainable AI (XAI): Integrating explainability tools like LIME and SHAP can improve prediction transparency and increase trust and clinical adoption among healthcare professionals.

4. Integration with IoT and Wearable Devices: Real-world patient data monitoring utilises wearable sensors. ML models in IoT-based health monitoring systems could forecast danger and alert for aberrant cardiac patterns in real time.

5: Deploy and Clinical Validation: Apply trained models as web-based or mobile applications in EHR systems. Model dependability, accuracy, and usability in real diagnostics require clinical trials and hospital validation.

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